

CPEN 416

Lecture 04: Projective measurements; introducing multi-qubit systems

Tuesday 22 September 2026

Announcements

- Quiz 2 today
- QHACK 2026 competition: 16-23 October (<https://qhack.ai>)



QHACK
QUANTUM CODING CHAMPIONSHIP 2026

YOUR SHOT AT THE BIGGEST PRIZE POOL IN QHACK HISTORY

OCTOBER 16-23, 2026 | The PennyLane-powered quantum computing coding competition, QHack, welcomes hobbyists, students, and curious developers to a hands-on challenge focused on accessible education and practical growth. Register now for access to exclusive resources to help you successfully compete.

Register now!

The image features a stylized illustration of a quantum atom with a central nucleus (a green sphere with a clock face) and three elliptical orbits (blue and yellow) with pink spheres at the ends. The entire graphic is set against a dark green background.

Last time

We played with *Pauli rotations*.

	Math	Matrix	Code	Special cases (θ)
RZ	$e^{-i\frac{\theta}{2}Z}$	$\begin{pmatrix} e^{-i\frac{\theta}{2}} & 0 \\ 0 & e^{i\frac{\theta}{2}} \end{pmatrix}$	<code>qp.RZ</code>	$Z(\pi), S(\pi/2), T(\pi/4)$
RY	$e^{-i\frac{\theta}{2}Y}$	$\begin{pmatrix} \cos \frac{\theta}{2} & -\sin \frac{\theta}{2} \\ \sin \frac{\theta}{2} & \cos \frac{\theta}{2} \end{pmatrix}$	<code>qp.RY</code>	$Y(\pi)$
RX	$e^{-i\frac{\theta}{2}X}$	$\begin{pmatrix} \cos \frac{\theta}{2} & -i \sin \frac{\theta}{2} \\ i \sin \frac{\theta}{2} & \cos \frac{\theta}{2} \end{pmatrix}$	<code>qp.RX</code>	$X(\pi), SX(\pi/2)$

Last time

We explored the effect of single-qubit unitary operations on the Bloch sphere.

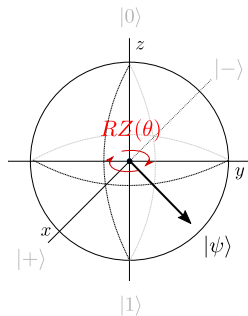
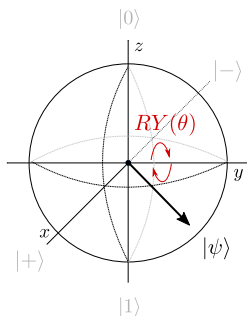
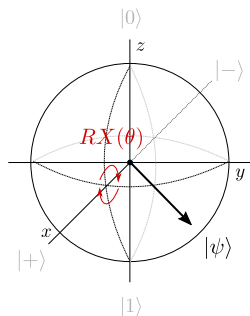


Image credit: Codebook node I.6

Last time

We introduced the “bra” part of bracket notation

$$|v\rangle = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} \quad \langle v| = (v_1^* \ v_2^*) = (|v\rangle)^\dagger$$

We defined the inner product to quantify *overlap* (similarity) between quantum states.

$$\langle v | w \rangle = (v_1^* \ v_2^*) \begin{pmatrix} w_1 \\ w_2 \end{pmatrix} = v_1^* w_1 + v_2^* w_2$$

Learning outcomes

- 1 Perform a projective measurement in the computational basis
- 2 Apply basis rotations to perform a projective measurement in any basis
- 3 Mathematically describe a system of multiple qubits
- 4 Describe the action of common multi-qubit gates

Inner products

np. vdot

Exercise from last time: compute the inner product between

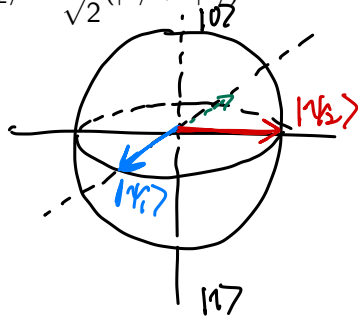
$$|\psi_1\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle), \quad |\psi_2\rangle = \frac{1}{\sqrt{2}}(|0\rangle + i|1\rangle)$$

Are they orthogonal?

Know: $|0\rangle, |1\rangle$ orthogonal

$|\psi_1\rangle = |+\rangle, |\psi_2\rangle$

$$\langle \psi_1 | \psi_2 \rangle = \frac{1}{\sqrt{2}} (1 \quad 1) \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix} = \frac{1}{2} (1 + i)$$



Inner products

Exercise from last time: compute the inner product between all possible combinations of $|0\rangle$ and $|1\rangle$.

$\langle 0 0\rangle$	1
$\langle 0 1\rangle$	0
$\langle 1 0\rangle$	0
$\langle 1 1\rangle$	1

$|0\rangle, |1\rangle$: computational basis

Orthonormal bases

For a single qubit, a pair of states that are **normalized** and **orthogonal** constitute an **orthonormal basis** for the Hilbert space.

Exercise: do the states

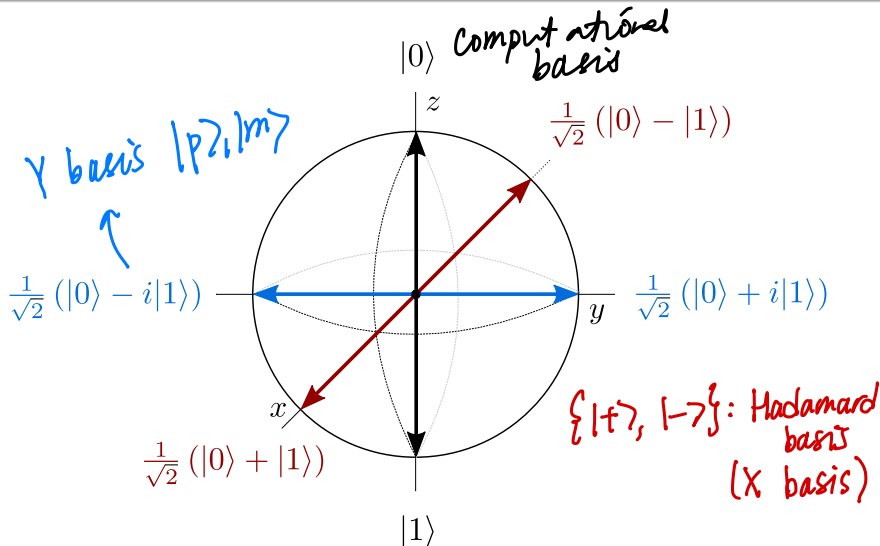
$$|p\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{i}{\sqrt{2}}|1\rangle, \quad |m\rangle = \frac{1}{\sqrt{2}}|0\rangle - \frac{i}{\sqrt{2}}|1\rangle$$

form an orthonormal basis? ✓

$\langle p p\rangle$	1
$\langle m p\rangle$	0
$\langle p m\rangle$	0
$\langle m m\rangle$	1

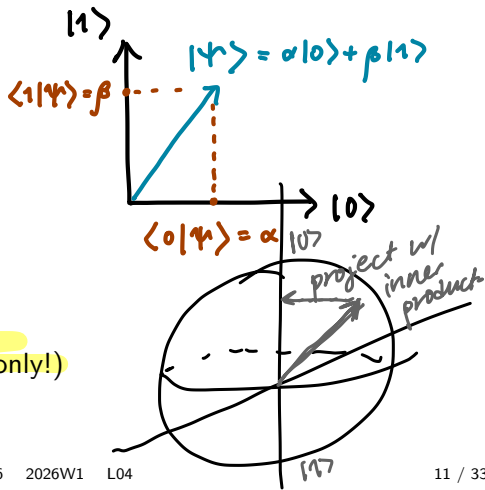
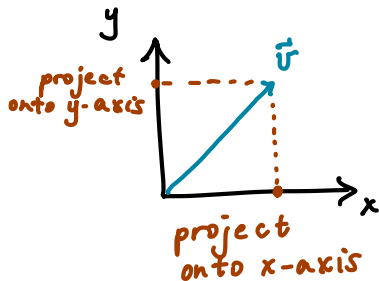
$$\begin{aligned} \langle p|p\rangle &= \frac{1}{\sqrt{2}}(1 - i) \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix} \\ &= \frac{1}{2}(1 - i^2) \\ &= 1 \\ \langle m|p\rangle &= \frac{1}{\sqrt{2}}(1 + i) \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix} \\ &= \frac{1}{2}(1 + i^2) \\ &= 0 \end{aligned}$$

Bases on the Bloch sphere



Projective measurements

Measurement is performed with respect to a basis; we perform **projections** to determine the overlap with a given basis state.



(Image for expository purposes only!)

Image credit: Xanadu Quantum Codebook 1.9

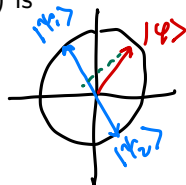
Projective measurements

When we measure a qubit in state $|\varphi\rangle$ with respect to basis $\{|\psi_i\rangle\}$, the probability of observing it in state $|\psi_i\rangle$ (outcome i) is

$$\text{Prob. outcome } i = |\langle\psi_i|\varphi\rangle|^2$$

(observe the qubit in $|\psi_i\rangle$)

If we observe outcome i , the qubit *remains* in $|\psi_i\rangle$.



Example: Let $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$. If we measure $|\psi\rangle$ in the computational basis,

$$\text{Prob } |0\rangle = |\alpha|^2 \Rightarrow |\langle 0|\psi\rangle|^2$$

$$\langle 0|\psi\rangle = \langle 0|(\alpha|0\rangle + \beta|1\rangle) = \alpha \underbrace{\langle 0|0\rangle}_{=1} + \beta \cancel{\langle 0|1\rangle} = \alpha$$

Measurement in the computational basis

Exercise: what are the measurement outcome probabilities for

$$|p\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{i}{\sqrt{2}}|1\rangle, \quad |m\rangle = \frac{1}{\sqrt{2}}|0\rangle - \frac{i}{\sqrt{2}}|1\rangle$$

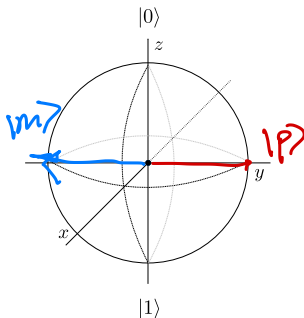
in the computational basis?

try yourself!

Measurement in the computational basis

We know these are *not* the same state: there is a relative phase.

$$|p\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{i}{\sqrt{2}}|1\rangle, \quad |m\rangle = \frac{1}{\sqrt{2}}|0\rangle - \frac{i}{\sqrt{2}}|1\rangle$$



How to tell them apart with a measurement?

Basis rotations

So far we've seen 3 types of measurements in PennyLane:

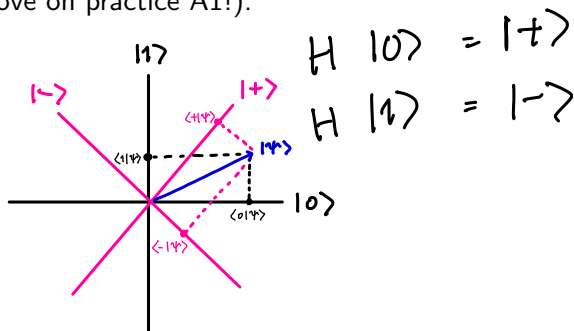
- 1 `qp.state()`
- 2 `qp.probs(wires=x)`
- 3 `qp.sample()`

But these all perform a computational basis measurement. This is also the only measurement allowed on most hardware.

How can we measure with respect to *different bases*?

Basis rotations

Unitary operations preserve length *and* angles between orthonormal quantum states (prove on practice A1!).



Projective measurements can be performed with respect to a different basis by applying a **basis rotation** to remap its states to computational basis states.

Image credit: Codebook node I.9

Basis rotations

Exercise: Suppose we want to measure in the “Y” basis. First, determine a sequence of gates that sends

$$|0\rangle \rightarrow |p\rangle = \frac{1}{\sqrt{2}}(|0\rangle + i|1\rangle)$$

$$|1\rangle \rightarrow |m\rangle = \frac{1}{\sqrt{2}}(|0\rangle - i|1\rangle)$$

$$\begin{array}{ll}
 |0\rangle \rightarrow \boxed{RX(-\pi/2)} \rightarrow |p\rangle & ? \quad |0\rangle \rightarrow \boxed{RY(\pi/2)} \rightarrow \boxed{RZ(\pi/4)} \rightarrow |p\rangle \\
 |1\rangle \rightarrow \boxed{RX(-\pi/2)} \rightarrow |m\rangle & ? \quad |0\rangle \rightarrow \boxed{H} \rightarrow \boxed{S} \rightarrow |p\rangle
 \end{array}$$

Basis rotations

At the end of our circuit, apply the reverse (adjoint) of this transformation to rotate *back* to the computational basis.

$$UU^\dagger = I$$



Let R : $R|0\rangle = |p\rangle$
 $R|1\rangle = |m\rangle$



If we measure and observe $|0\rangle$, we know the qubit was previously $|p\rangle$ in the Y basis (analogous result for $|m\rangle$).

$$|\psi\rangle = \alpha |p\rangle + \beta |m\rangle \Rightarrow R^\dagger |\psi\rangle = \alpha R^\dagger |p\rangle + \beta R^\dagger |m\rangle = \alpha |0\rangle + \beta |1\rangle$$

Adjoint

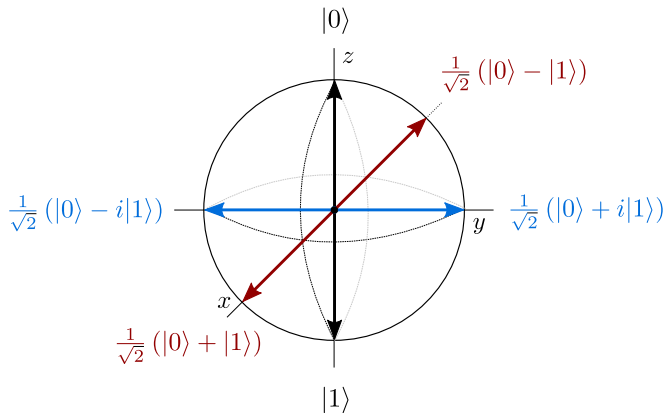
PennyLane can compute adjoints of operations *and* entire quantum functions using `qp.adjoint`:

```
def some_function(phi):  
    qp.RX(phi, wires=0)  
  
def apply_adjoint(phi):  
    qp.adjoint(qp.S)(wires=0)  
    qp.adjoint(some_function)(phi)
```

`qp.adjoint` is a special type of function called a **transform**.

Projective measurements and the Bloch sphere

Projective measurement corresponds to a geometric projection onto the axis represented by a basis.



H - RZ exercise, revisited

Exercise: suppose we run the following circuit, but perform the measurement in the Y basis.



- 1 Using the Bloch sphere as intuition, what do you think the outcome probability of $|p\rangle$ looks like as a function of θ ? Discuss with a neighbour.
- 2 Test your predictions in software.

Next time

Content:

- Entanglement
- Measurement of multi-qubit states
- Bell basis measurements

Start w/ mathematics of multi-qubit systems

Action items:

- ① Work through last section of yesterday's tutorial (bases)
- ② Practice assignment 1

Recommended reading:

- For today: Codebook IQC, SQ, MQ; N & C 1.2, 1.3.1-1.3.5, 2.2.3-2.2.5, 4.3
- For next time: Codebook SQ MQ (all tied up); N & C 1.3.6, 1.3.7, 2.3